



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

CHEMISTRY
PAPER 1

9189/1

NOVEMBER 2017 SESSION

2 hours

Additional materials:

Answer paper Graph paper
Data Booklet
Mathematical tables and/or electronic calculator

TIME: 2 hours

INSTRUCTIONS TO CANDIDATES

Write your name, Centre number and candidate number in the spaces provided on the answer paper/answer booklet.

Answer **six** questions.

Answer **two** questions from Section A, **one** question from Section B, **two** questions from Section C and **one** other question chosen from any section.

Write your answers on the separate answer paper provided.

If you use more than one sheet of paper, fasten the sheets together.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

You are reminded of the need for good English and clear presentation in your answers.

This question paper consists of 7 printed pages and 1 blank page.

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Section A

Answer at least **two** questions from this section.

- 1 (a) (i) Explain the trend in boiling points of noble gases with increase in atomic number.
- (ii) Draw a dot and cross diagram to show the bonding in SO_3^{2-} .
- (iii) State the shape of SO_3^{2-} . [3]
- (b) Explain the following observations:
- (i) the boiling point of water is higher than that of hydrogen sulphide
- (ii) graphite is used as a high temperature lubricant
- (iii) CO_2 is a gas at room temperature but SiO_2 is a solid at room temperature [6]
- (c) A current of 0.200 A is passed through a molten nitrate, $\text{M}(\text{NO}_3)_n$ for 3 hours. The metal, M, has a relative atomic mass 27.
- Calculate the charge of the ions of M, if 0.201 g of M were deposited on the cathode. [3]
- [Total: 12]

- 2 The equation shows a dynamic equilibrium at r.t.p.



- (a) (i) Define the term *dynamic equilibrium*.
- (ii) Calculate the equilibrium constant, K_c , given that 50% of N_2O_4 had dissociated at equilibrium. [4]
- (b) (i) Explain why a catalyst does not affect the position of equilibrium. [1]
- (ii) State, giving reasons, how an increase in temperature would affect the
- equilibrium composition,
 - K_c value.
- [5]

- (c) Calculate the enthalpy change for the reaction, $\Delta H_r^\ominus$, given that $\Delta H_f^\ominus(\text{N}_2\text{O}_4) = +9.70 \text{ kJmol}^{-1}$ and $\Delta H_f^\ominus(\text{NO}_2) = +33.9 \text{ kJmol}^{-1}$. [2]

[Total: 12]

- 3 (a) (i) Define the term *solubility product*.

- (ii) State the limitation of the use of solubility product.

[2]

- (b) (i) Derive an expression for the solubility product of $\text{Mg}_3(\text{PO}_4)_2$.

- (ii) Given that the solubility of $\text{Mg}_3(\text{PO}_4)_2$ is $1.0 \times 10^{-2} \text{ moldm}^{-3}$, calculate the solubility

1. product of $\text{Mg}_3(\text{PO}_4)_2$,
2. of $\text{Mg}_3(\text{PO}_4)_2$ in $0.1 \text{ moldm}^{-3} \text{ Na}_3\text{PO}_4$
3. of $\text{Mg}_3(\text{PO}_4)_2$ in $0.1 \text{ moldm}^{-3} \text{ MgSO}_4$.

[5]

- (c) (i) The enthalpies of formation, $\Delta H_f^\ominus$, of $\text{CH}_{4(\text{g})}$, $\text{H}_2\text{O}_{(\text{g})}$ and $\text{CO}_{(\text{g})}$ at 298 K are -75 kJmol^{-1} , -242 kJmol^{-1} and -110 kJmol^{-1} respectively.

Calculate the $\Delta H_r^\ominus$ for the reaction



- (ii) Describe the effect of increasing

1. temperature and
2. pressure on the reaction in (i).

- (iii) Suggest **one** use of any of the products of the reaction in (i).

[5]

[Total: 12]

Section B

Answer at least one question from this section.

4 (a) State the structure and bonding in carbon and lead. [2]

(b) Compare and contrast the

- (i) structure and bonding of CO_2 and PbO_2 ,
- (ii) relative stability of the lower and higher oxidation states of carbon and lead,
- (iii) thermal stability of the tetrachlorides of carbon and lead,
- (iv) hydrolysis of the tetrachlorides of carbon and lead.

[8]

(c) Sn and Pb show the inert pair effect.

Explain the inert *pair effect*.

[2]

[Total: 12]

5 (a) State, with a reason, the

- (i) most electronegative halogen,
- (ii) halogen that is most likely to have a positive oxidation state.

[4]

(b) Write equations to show how each of the following substances react with concentrated sulphuric acid:

- (i) sodium fluoride,
- (ii) sodium iodide

[3]

(c) Describe the observations made when

- (i) potassium permanganate crystals are mixed with concentrated hydrochloric acid,
- (ii) aqueous bromine is mixed with an excess of potassium iodide and aqueous sodium thiosulphate.

[5]

[Total: 12]

Section C

Answer at least **two** questions from this section.

- 6 Fig. 6.1 shows the structural formula of an organic compound.

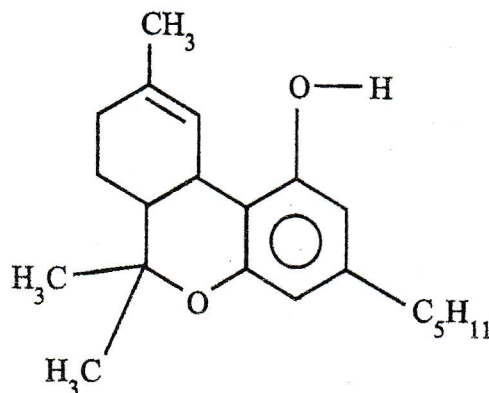


Fig. 6.1

- (a) Name any **two** functional groups presents in the compound. [2]
- (b) Give the structure of the organic product formed when the compound is reacted with
- (i) Br_2 in an inert solvent,
 - (ii) hot concentrated KMnO_4 ,
 - (iii) Na solid,
 - (iv) CH_3COCl
 - (v) benzenediazonium salt.
- [5]
- (c) State the observations made in each of the reactions in (b). [5]
- [Total: 12]

- 7 (a) (i) Define the term *nucleophilic substitution*.
 (ii) State the properties of halogenoalkanes which makes them undergo nucleophilic substitution. [2]
- (b) State the reagent(s) and conditions for the conversion of 1-bromopropane to
 (i) propene,
 (ii) butanenitrile. [4]
- (c) Fig. 7.1 shows a reaction scheme for the conversion of an organic compound A, $C_4H_{10}O$, to organic compounds C and D.

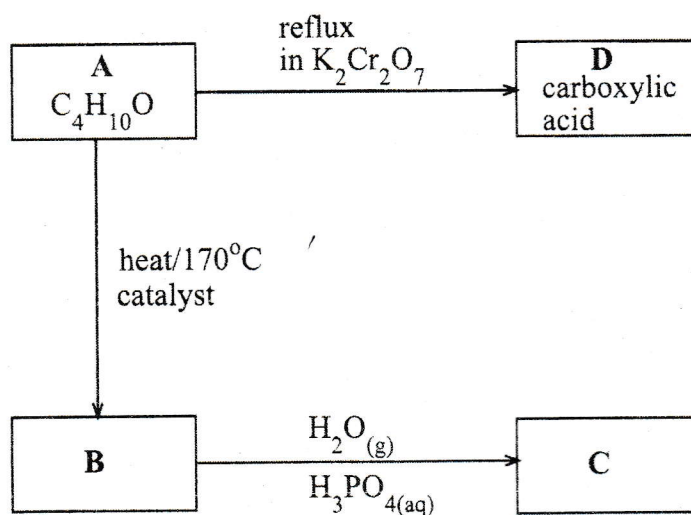


Fig. 7.1

- Deduce the structural formulae of A, B and C, given that A and C are isomers and C is resistant to oxidation. [3]
- (d) Name the type of reaction occurring in the conversion of
 (i) A to D,
 (ii) A to B,
 (iii) B to C.

[3]
 [Total: 12]

- 8 (a) Compare and contrast the reactions of pentanol and phenol with each of the following reagents:
- (i) $\text{NaOH}_{(\text{aq})}$
 - (ii) Na_2CO_3
 - (iii) $\text{Br}_{2(\text{aq})}$
 - (iv) ethanoic acid
- [7]
- (b) Explain why
- (i) methylbenzene is more readily nitrated than benzene,
 - (ii) nitrobenzene is less readily nitrated than benzene.
- [2]
- (c) State and explain the differences in the acidities of ethanol and phenol. [3]
- [Total: 12]